NETFLOW

YERLİ ÜRETİM

LIQUID LEVEL MEASUREMENT · VSG SERIES

# MAGNETIC LEVEL INDICATOR

By-pass type analog level indicator for tanks and vessels, with no contact to the medium

The magnetic by-pass level indicator allows the liquid level in tanks to be monitored from outside by means of a magnetic float and a moving colour-changing indicator.

| | |
|---|---|
| **PN10**<br>10 BAR | **100 °C / 250 °C**<br>MAX. OPERATING TEMPERATURE |
| **5800 mm**<br>ONE-PIECE LENGTH | **AISI 304 / 316**<br>FULLY STAINLESS STEEL |



VSG STANDARD

## FEATURES

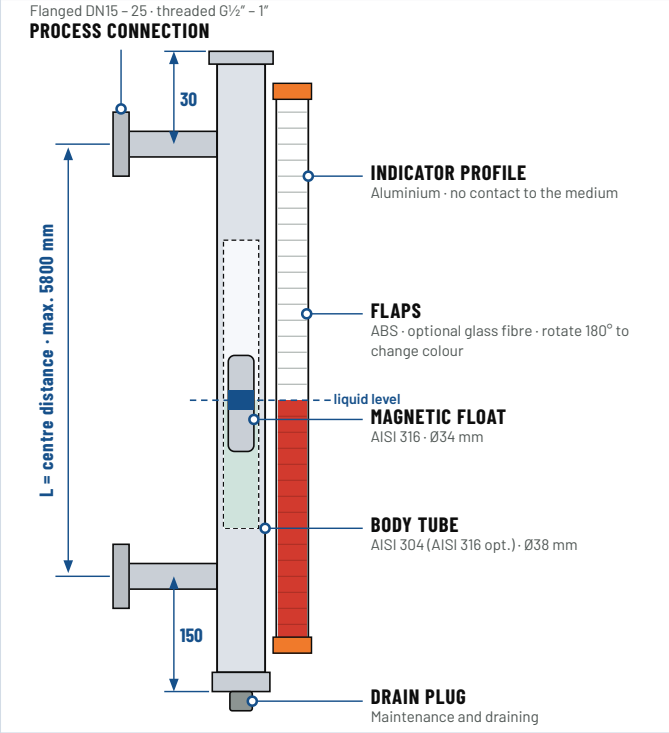
- Indicator profile with no contact to the medium
- Analog indicator easily readable from a distance
- Robust, shock-resistant body
- Dry contacts at any point with limit switches mounted on the indicator
- Easy installation with standard and swivel flanges
- Easy maintenance and installation with the valved model
- Custom manufacturing for different connections and lengths
- Level reading on the device or in percent, volume or metres with the scaled model

## TECHNICAL SPECIFICATIONS

| | |
|---|---|
| Pressure (Pmax) | PN10 |
| Temperature (Tmax) | 100 °C · optional max. 250 °C |
| Connection material | Carbon steel St 37.2 · stainless steel (optional) |
| Connection types | Threaded G½" · G¾" · G1"<br>Flanged DN15 · DN20 · DN25 |
| Max. length | 5800 mm, one piece |

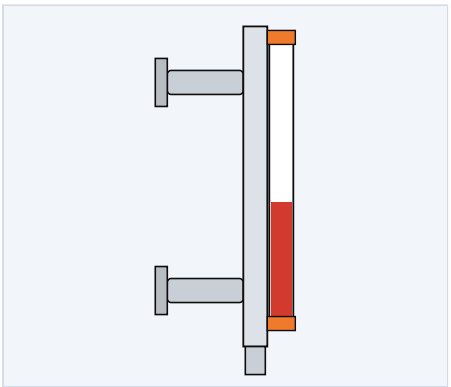
| | |
|---|---|
| Float | AISI 316 · Ø34 mm |
| Tube | AISI 304 (AISI 316 optional) · Ø38 mm |
| Indicator profile | Aluminium |
| Flaps | ABS plastic · optional glass fibre (200 °C resistant) |
| Contact / signal | Dry contact via limit switch · 0–10 V / 4–20 mA (VSG 30) |

## DESIGN, OPERATING PRINCIPLE AND DIMENSIONS



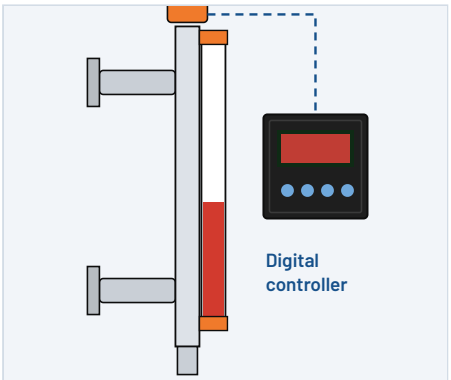

1. **Communicating vessels.** The indicator is connected to the tank as a by-pass via two process connections. The liquid level in the body tube is the same as in the tank.
2. **Magnetic float.** The float inside the tube floats on the liquid surface and moves with the level.
3. **Flap indicator.** The magnet in the float rotates the flaps in the profile by 180°. Below the level they show red, above white; the level can be read from a distance.
4. **Contact and signal.** Limit switches mounted on the profile provide dry contacts at the required points. In VSG 30 models the level is transmitted as an analog signal.

## MODELS



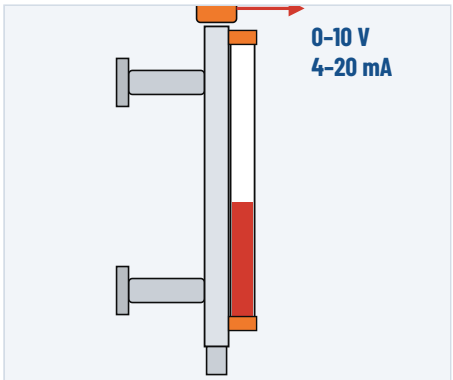
### VSG STANDARD

Mounted on the tank or vessel as a by-pass. Level changes are monitored visually via the flaps. Limit switches can be fitted.




### VSG 30

The level signal is transmitted to a digital controller, which displays the level in percent, volume, mass or metres.



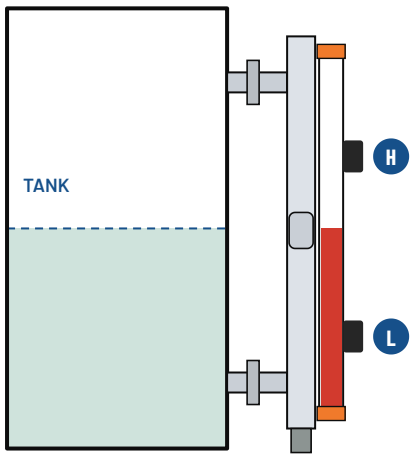

### VSG 30 DIRECT OUTPUT

The level transmitter is integrated into the indicator. The level is sent directly to the automation system as a 0–10 V or 4–20 mA signal.

## ACCESSORIES

| ACCESSORY | DESCRIPTION | CONTACT / OUTPUT | VOLTAGE | TEMPERATURE |
|---|---|---|---|---|
| **Limit switch** | Mounted on the indicator profile, provides a dry contact at the required point. Connected via controller or directly to automation. Aluminium housing. | 1.5 A · 50 W max.<br>NO · NC (opt.) | 250 VAC – 200 VDC | –60 … +125 °C |
| **High-temperature switch** | Mounted on the indicator profile in high-temperature applications, provides a dry contact. Controller supported. Aluminium housing. | 1.5 A · 50 W max.<br>NO · NC (opt.) | 250 VAC – 200 VDC | –60 … +250 °C |
| **Controller** | Switches between the limit switches and the devices to be controlled. Up to 6 limit switches can be connected. | 6 contact inputs<br>3 × 10 A NO + NC | 220 VAC 50 Hz<br>24 VDC (opt.) | – |
| **Digital controller** | Used with VSG 11. The level is shown on the display and transmitted to automation via relay and analog outputs. | 3 × 10 A NO + NC<br>4–20 mA · 0–10 V | 220 VAC 50 Hz<br>24 VDC (opt.) | – |
| **Analog indicator** | Connected to VSG 11 or a digital controller; the liquid level is shown on the indicator display. | – | 24 V | – |

(opt.) = optional




H / L: limit switch points

## INSTALLATION AND USE

**01** The indicator is mounted vertically on two process connections on the tank side wall using standard or swivel flanges or threaded connections.

**02** The centre distance (L) is specified in the order according to the tank connections; manufactured in one piece up to 5800 mm.

**03** With the valved model, the process valves can be closed and the indicator serviced without draining the tank.

**04** Limit switches are moved to the required level on the indicator profile, fixed, and provide dry contacts.

**05** Sediment and residue in the tube are drained periodically through the drain plug.

**CAUTION** There must be no strong magnetic field or magnetic material near the indicator; it may affect correct flap rotation.

**NETFLOW** Custom manufacturing for other connections, lengths, pressures and temperatures. Orders and technical support: +90 541 373 79 53 · info@netflw.com